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SEAT No. :

**PD41**

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**[6409]-242**

**S.E. (AI & ML) (Insem)**

**FUNDAMENTALS OF AI & ML**

**(2019 Pattern) (Semester-IV) (218553)**

*Time : 1 Hour]*

*[Max. Marks : 30*

*Instructions to the candidates:*

- 1) *Attempt Q.1 or Q.2 Q.3 or Q.4.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume suitable data, if necessary.*

**Q1)** a) What do mean by Intelligent Agent? Explain the structure of agent in environment with suitable diagram? [6]

b) What is PEAS? [4]

c) Differentiate between Artificial Intelligent and Machine Learning. [5]

OR

**Q2)** a) What is AI problem? Describe any two AI problems. [5]

b) What is meant by AI Turing test? Explain with suitable example. [5]

c) What is the evolution of artificial intelligence (AI) over time, and what are some of the major applications of AI in various fields and industries?[5]

**Q3)** a) Explain A\* Search Algorithm with example? State advantages and disadvantages of A\* Search Algorithm. [8]

b) What is difference between search algorithm and Local search algorithm? [7]

OR

**Q4)** a) What is constraint satisfaction problem? Explain back tracking search for CSPs? [7]

b) What is Best First Search Algorithm? What are the advantages of Depth First Search? [8]

